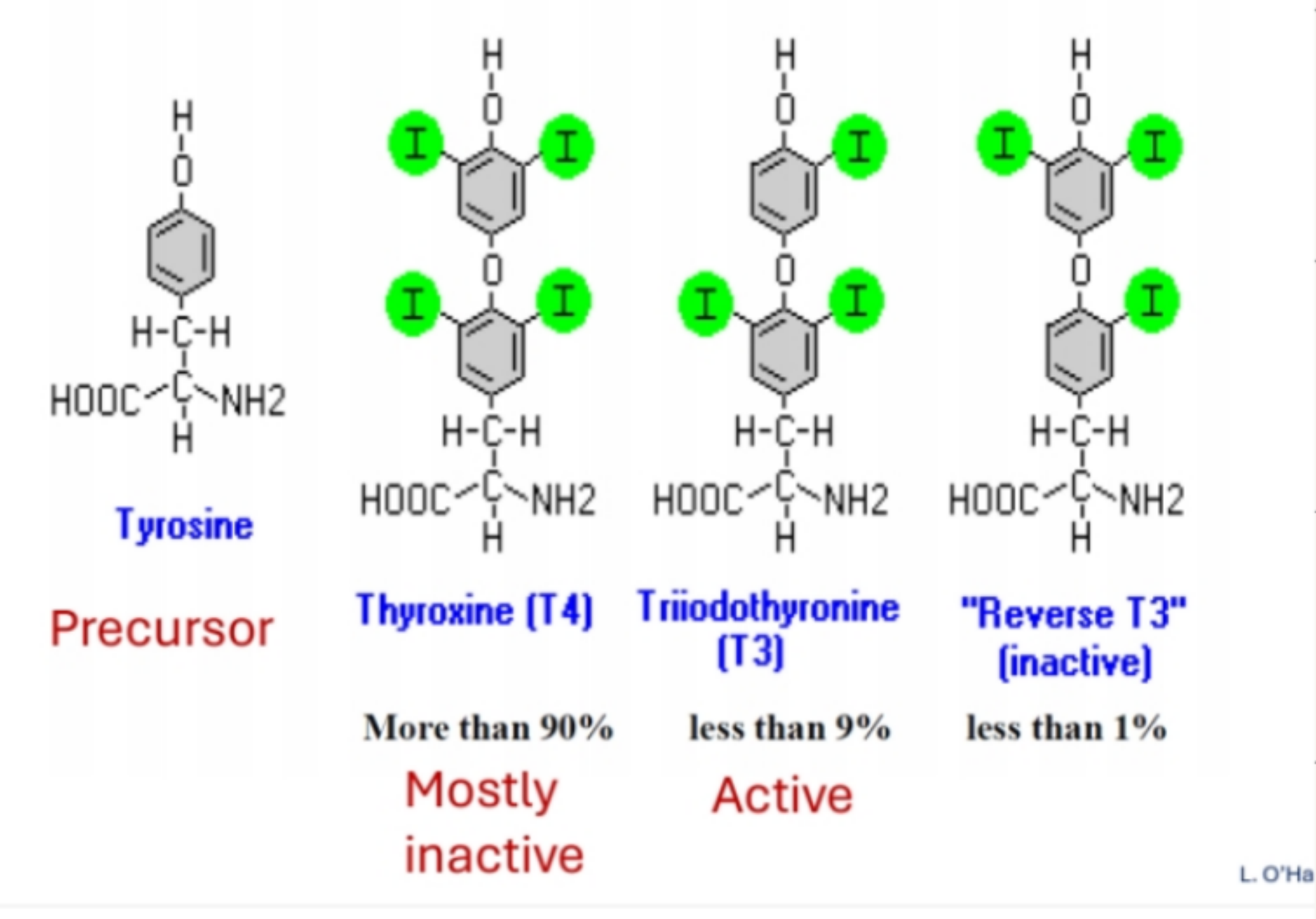
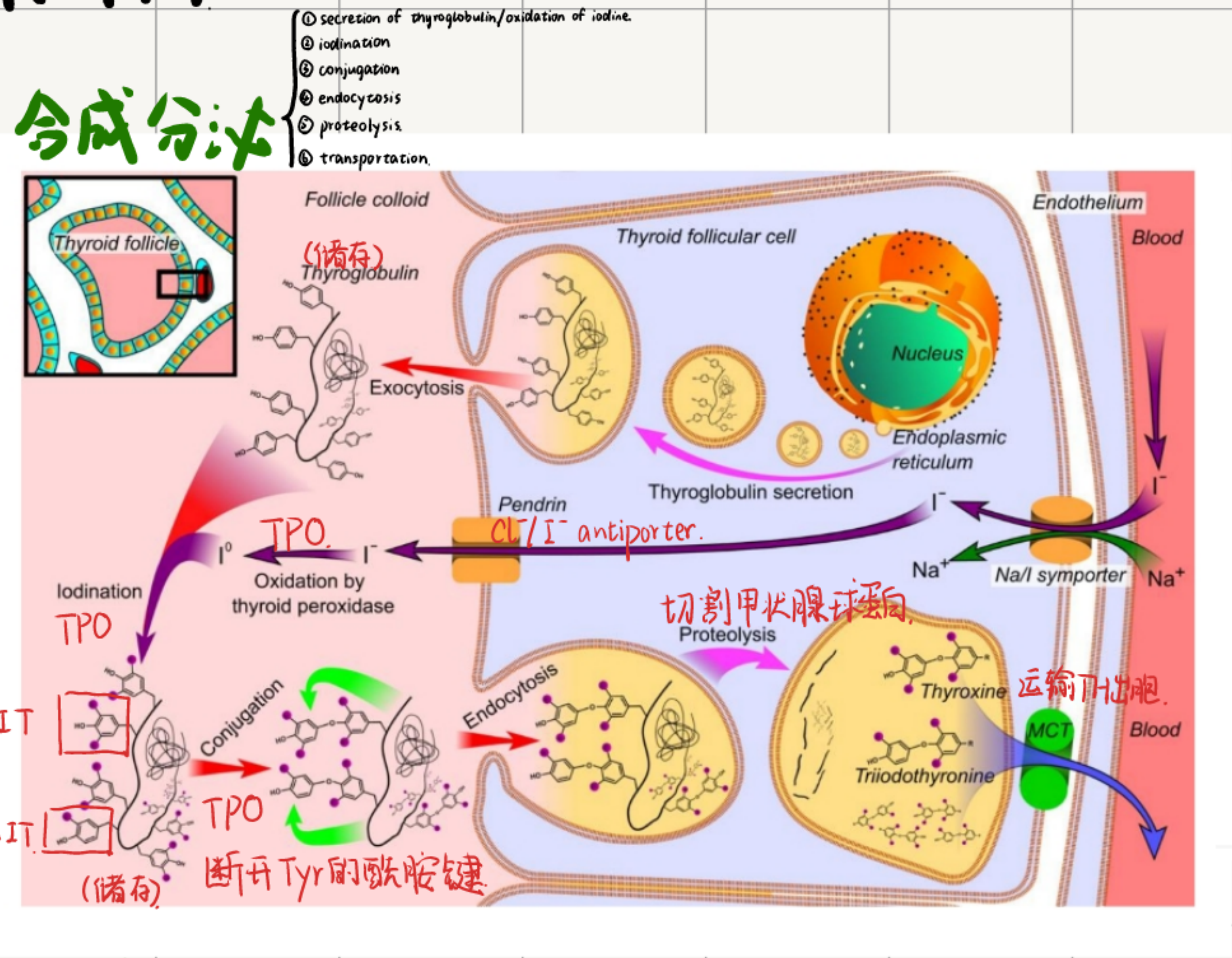


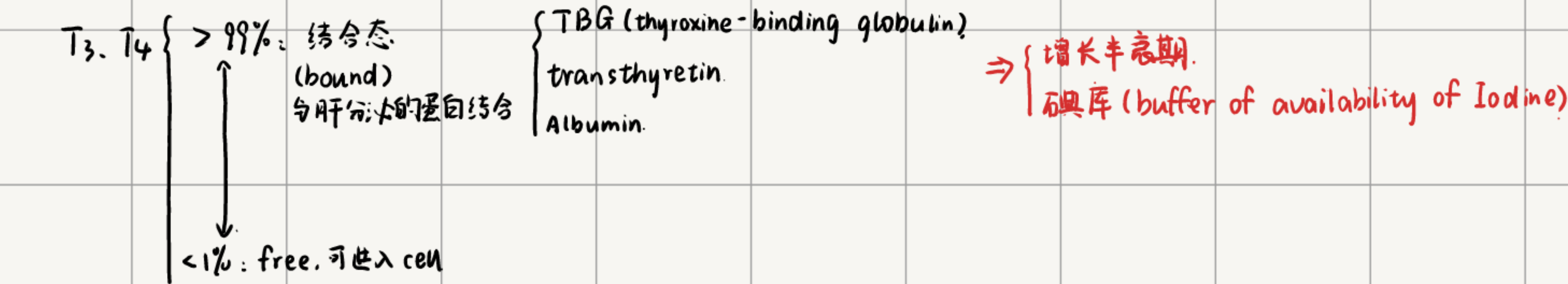
消化系统调节: TH, 肠相关激素, 脑肠轴

1. TH

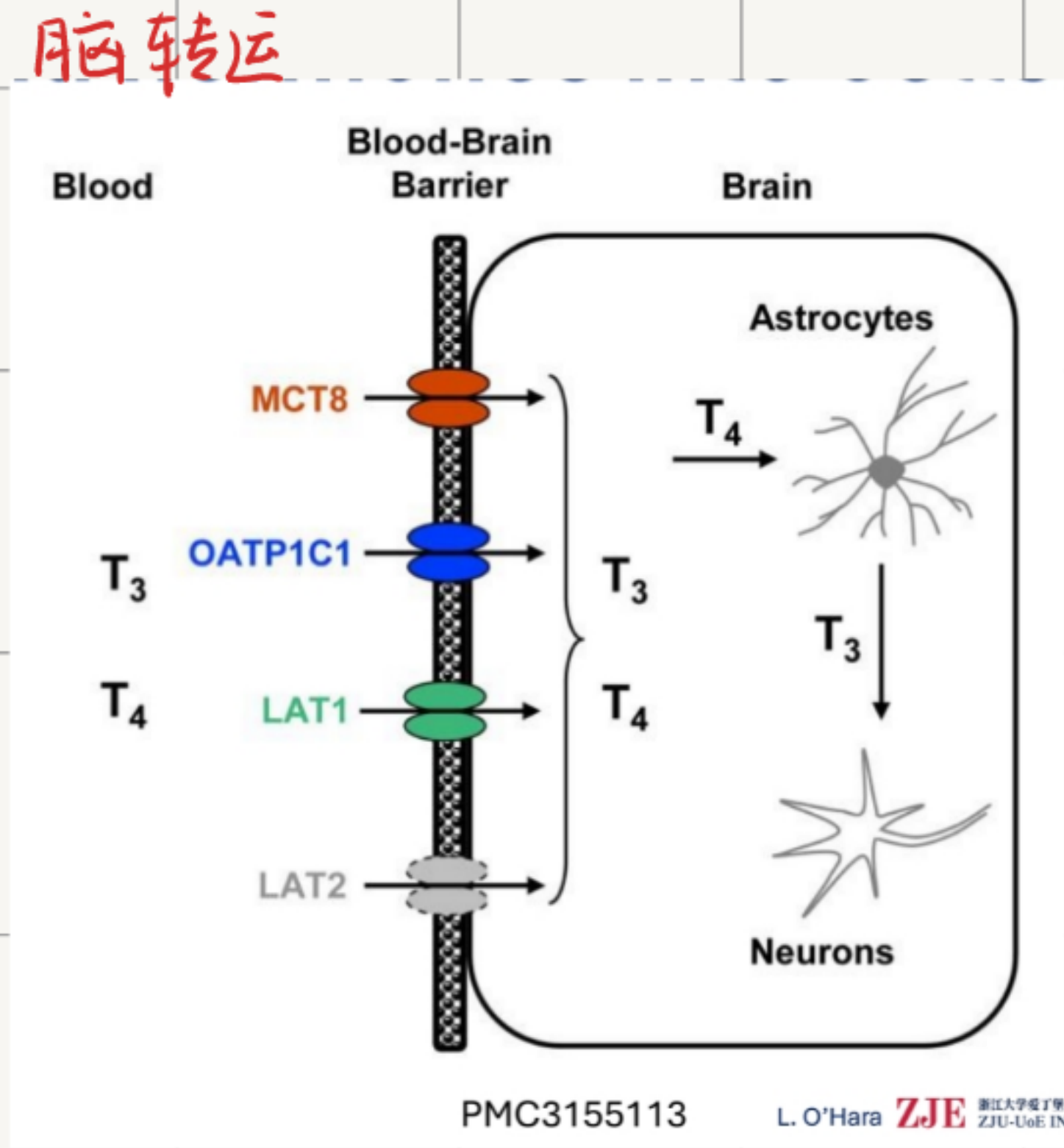
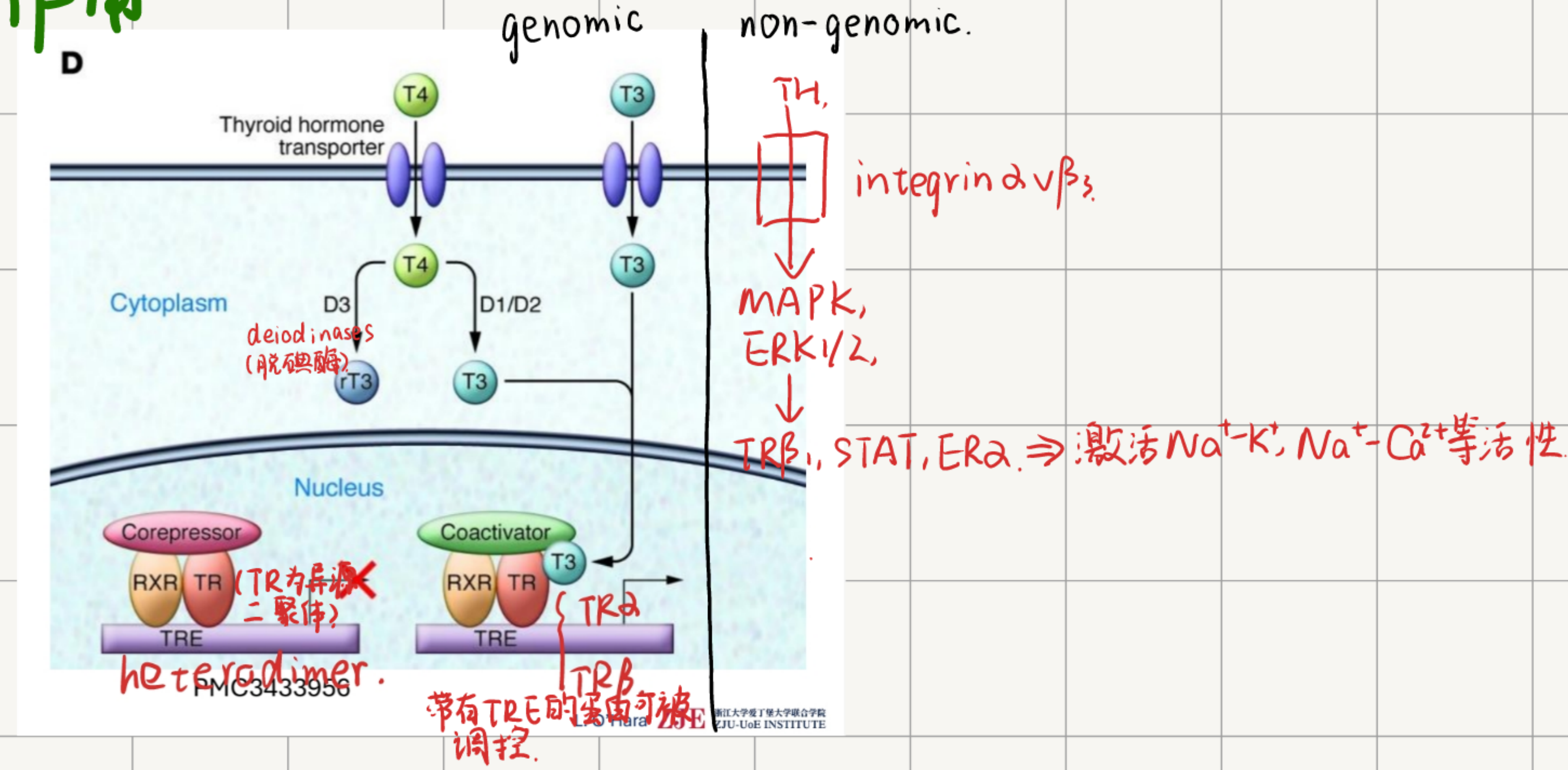
合成分泌



运输



作用



- ① increase thermogenesis: 增强BAT中UCP表达
- ② increase basal metabolic rate: 提高插入的 $\text{Na}^+\text{-K}^+$ 泵量, 加大ATP消耗.
- ③ increase availability of substrates for respiration.

糖: gluconeogenesis↑, insulin↑, 食欲↑.

脂: { lipolysis (adipose)↑, β -oxidation (mt)↑
lipogenesis (liver)↑.

将LDL从血流除去

④ increase β receptor expression: TH允许了肾上腺素的作用.

⑤ fetal development:

骨: 促 osteoblast proliferation, 以及骨形成过程中的基因表达.
脑: myelination, synaptogenesis.

permissiveness

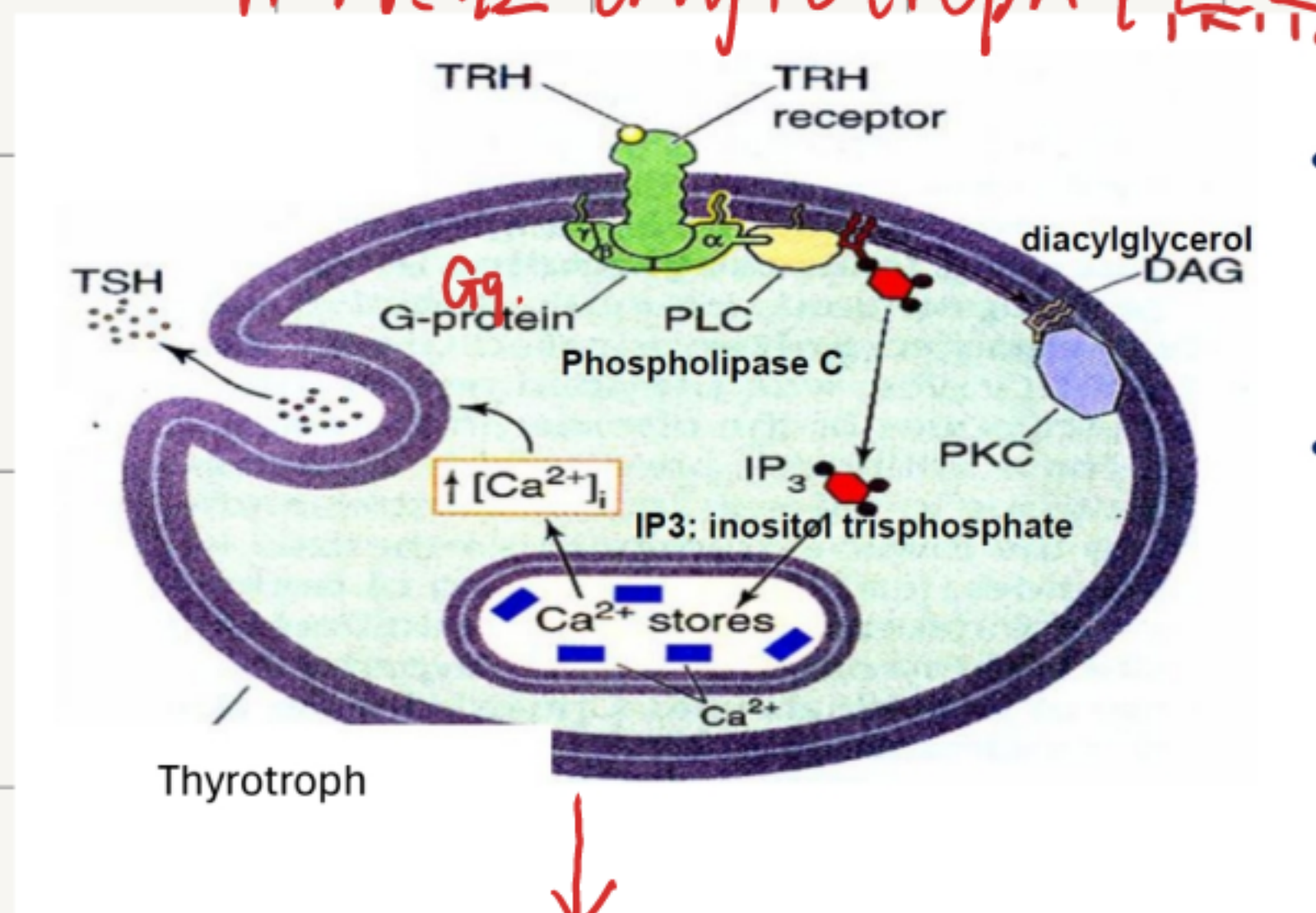
调节

parvocellular neurons in
paraventricular nucleus
(Hypothalamus)

← cold, exercise

— T_3 , T_4

TRH: 在 thyrotroph 中通过 Gq 促 TSH. (anterior pituitary)



TSH: 与 LH, FSH 有相同的亚基

结合 Gq/Gs GPCR TSHR 后: { G_s { 转录↑: TPO, thyroglobulin, Na^+-I^- symporter.

磷酸化: Na^+-I^- symporter, $Cl^- - I^-$ antiporter 转位至膜.
(pendrin)

Gq: 胞吞, 胞吐

促进 TH 入血

相关疾病: 因 T_4 含量高, 以 T_4 作为测量标准

hyperthyroidism: Grave's disease (自身抗体激活 TSHR)

治疗: β 抑制剂, TPO 抑制

hypothyroidism { primary { 缺 I: Goitre, Dwarfism

Hashimoto's thyroiditis:
自身抗体进攻甲状腺.

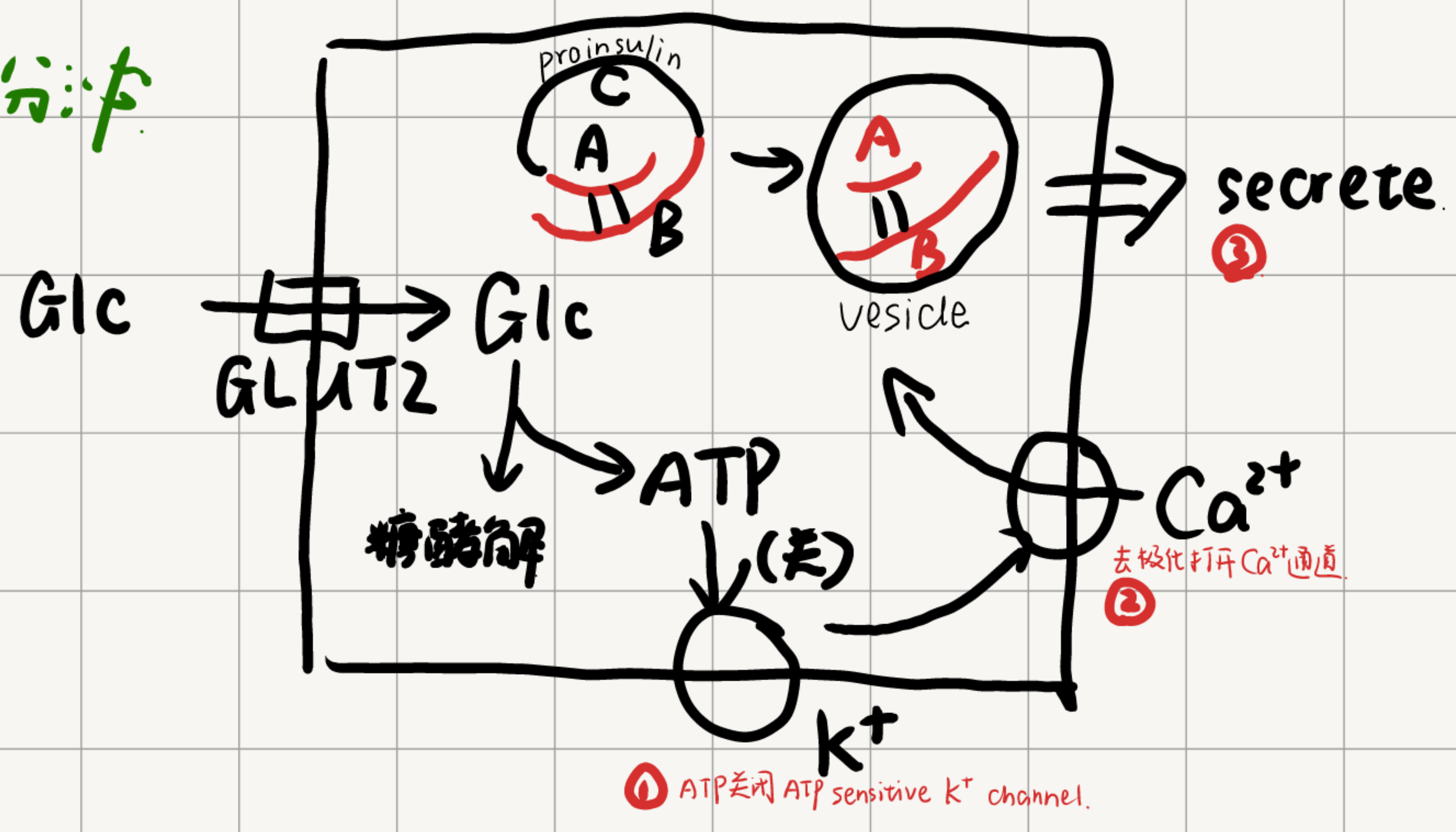
secondary: pituitary tumor 导致低 TSH.

2. 胰岛内分泌

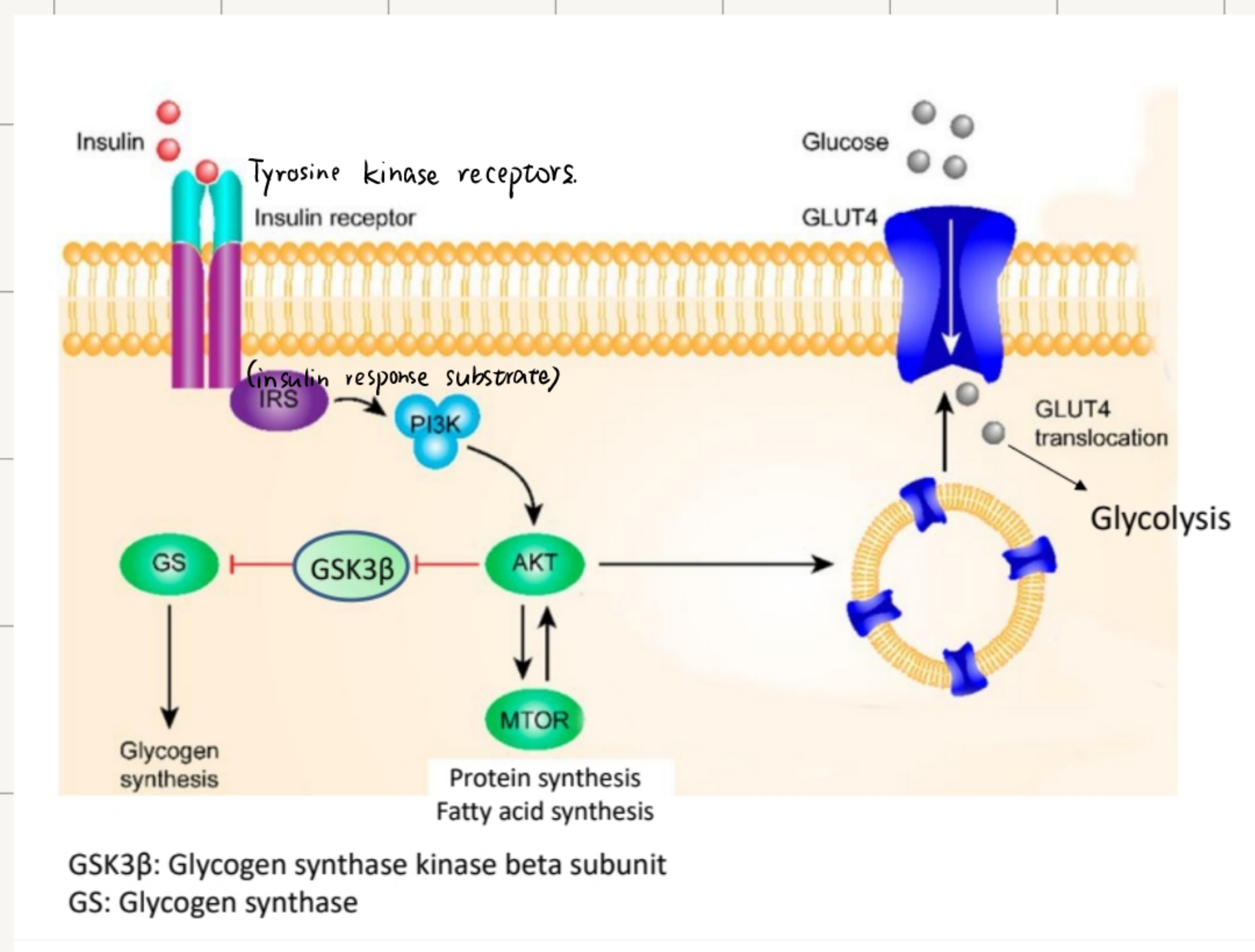
结构: pancreas {
exocrine {
 acinar cell: 分泌消化酶
 duct cell: 分泌 aqueous alkaline
endocrine: islet of Langerhans.

① β cell: insulin

合成分泌



信号转导:



通过 Tyr 激酶 激活 Akt {
 → GSK3 β → 糖原合成
 → mTOR → 蛋白质合成
 → GLUT4 转位, 吸收 Glc.

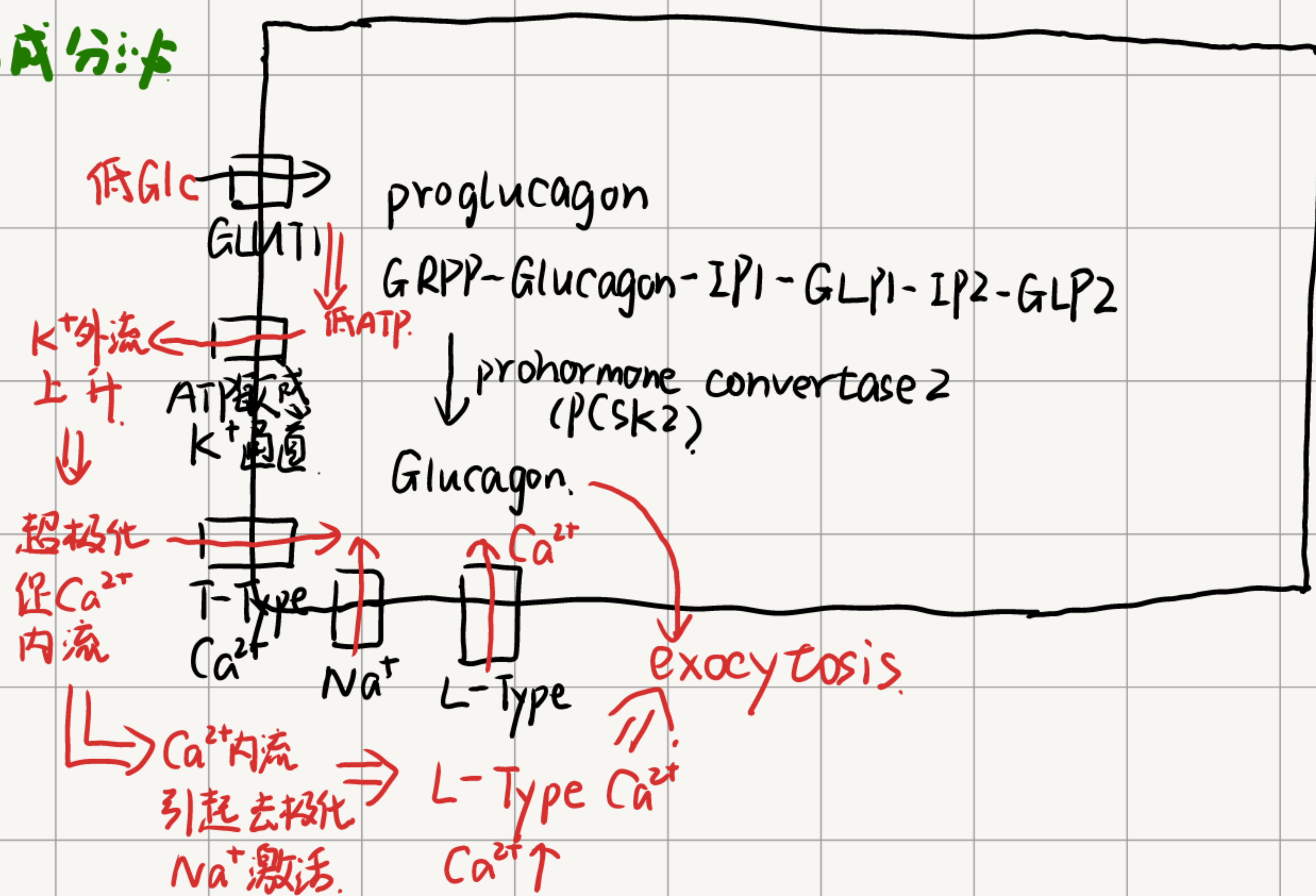
作用: ① 糖类代谢: {
 促 glycogenesis (糖原合成), glycolysis (糖酵解), glucose uptake
 抑 glucose conversion, glycogenolysis (糖原降解), gluconeogenesis (糖异生)

- 肝: 糖的储存 ↑
- 肌: 糖吸收 ↑
- 脂: 脂肪合成 ↑
- 脑: 进食中接受 insulin 调控 (satiety center in hypothalamus)

③ 降低血 aa 水平, 提高蛋白合成.

交感↑/肾上腺素↑ $\Rightarrow$ insulin↓

合成分泌



Glucagon $\Rightarrow \begin{cases} G_s \\ G_q \end{cases} \Rightarrow \begin{matrix} \text{糖原分解} \\ \text{糖异生} \end{matrix}$

①糖: 促糖原降解. 促 $G6P \rightarrow Glc$. 促糖异生.
抑糖原合成.

② 脂: 促进脂肪降解. 促 ketone 肝合成 (脑可用酮酸, 为脑提供资源)

③ 蛋白: 促进肝中蛋白降解.

对血中 aa: 不影响. 肝中降解蛋白产生 aa 转为 Glc

对肌蛋白: { 正常情况下不影响.
饥饿时肌蛋白↓.

调控:

① Glc↓ 促 glucagon (主要)

② insulin, somatostatin ↑ 抑 glucagon

③ 副交感 抑 glucagon

④ aa: 摄入高蛋白后: { insulin ↑ ⇒ 促蛋白合成.
glucagon ↑ ⇒ 促 aa 转化为 Glc.

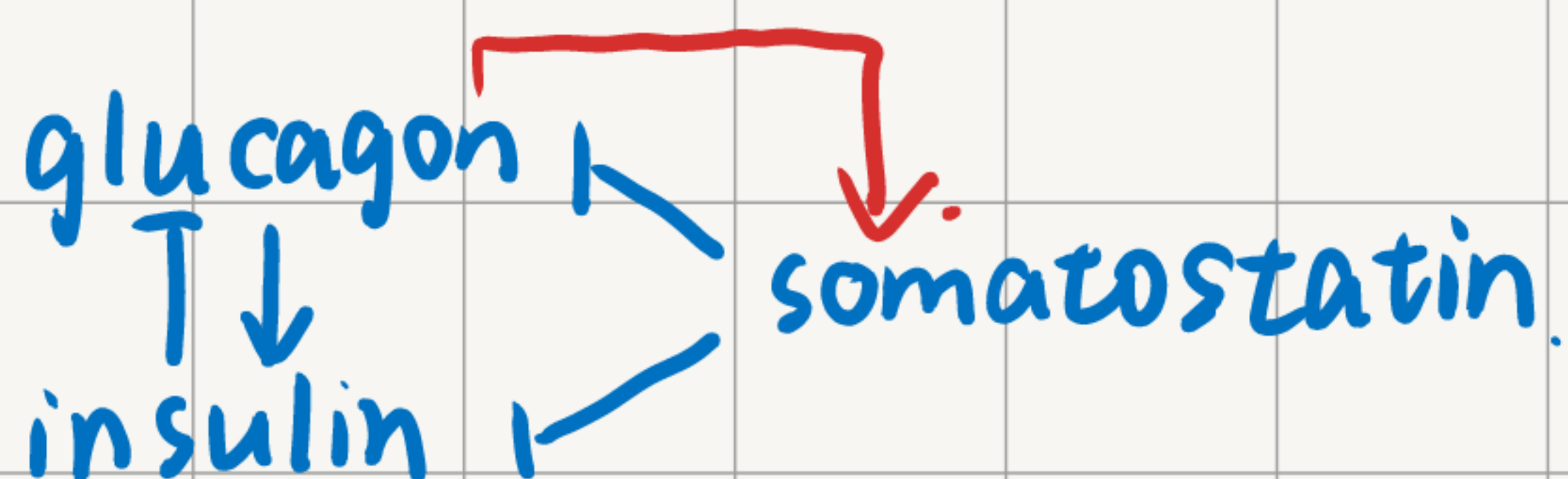
③ somatostatin: delta 细胞分泌.

在 hypothalamus 中分泌可抑 GH.

在 { delta cell 分泌可抑消化. insulin, glucagon.
| D cell

⇒ 延长食物逗留时间.

受 Glc, FA, aa ↑ 而分泌 (同 insulin)



3. 脑肠轴.

① glucose sensor { 感受 Glc { GPCR: TIR2, TIR3 $\Rightarrow$ Gq
GLUT/SGLT.

brain: taste cell (TIR2/TIR3)

gut: EEC (SGLT 通过类似胰岛素的方式引发 Ca^{2+} ↑,

导致 GLP 胞吐). neuropod: 下游与迷走神经形成突触

portal vein: primacy Glc sensor

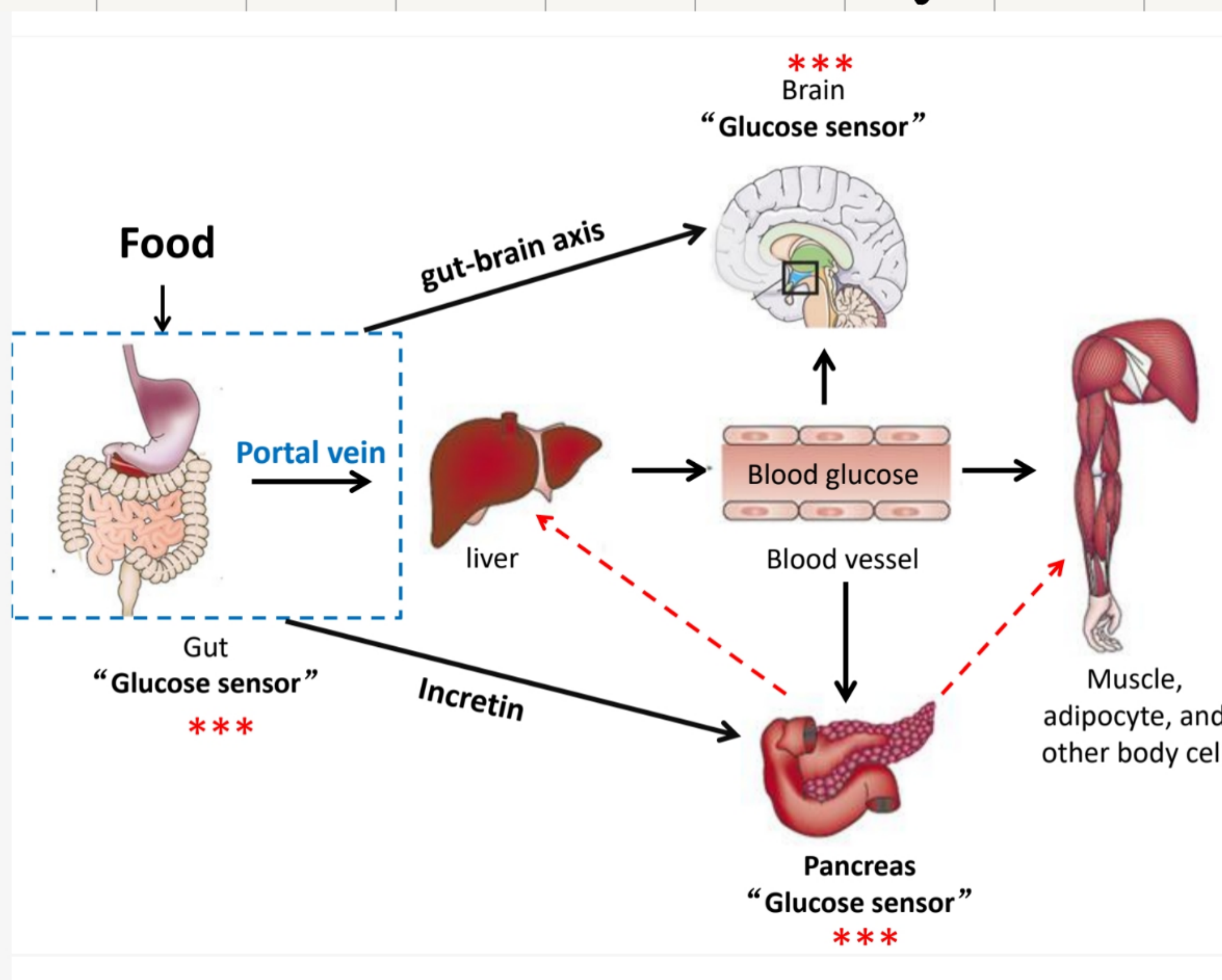
endothelial $\rightarrow$ { vagal } $\rightarrow$ 行为改变, 减少糖摄入
spinal

门静脉感受器的意义:

① more sensitive than systematic

② rule out intestinal consumption

③ consider gut gluconeogenesis.



② 脂质感受: long chain fatty acid 进入 EEC 后引发 PKC 激活, 释放 GLP-1 和 CCK.

③ 蛋白 sensor { aa: 结合后通过 CaSR. GPCR 激活 GLP, CCK.
寡肽: 通过 Pept1 通道进入 cell →

④ 食欲: ^{satiety center (hypothalamic arcuate)} { NPY/Agrp: orexigenic
POMC: anorexigenic ← leptin (adipose tissue)

4. incretin 和 diabetes

incretin:

PSCK2 (α cell) → glucagon

合成分泌: proglucagon $\xrightarrow{\text{PSCK (肠. A)}} \text{GLP-1, GLP-2, IP}$

Hormone	Amino acids (active form)	Cell type	Location	Major stimulus	Major action	Receptor
GLP-1 <i>glucagon-like peptide</i>	37	L-cell	Especially the terminal ileum*	High intestinal glucose/nutrients	Incretin #	GPCR
GIP <i>gastric inhibitory polypeptide</i>	42	K-cell	Duodenum and jejunum	High intestinal glucose/nutrients	Incretin #	GPCR

信号转导: $\text{GLP} \rightarrow \text{GLP-1R} \rightarrow \text{Gs} \rightarrow \text{cAMP} \rightarrow \text{PKA} \rightarrow \text{电位改变} \rightarrow \text{insulin}$

作用: ① pancreas: { 促 insulin 分泌
GIP 促 glucagon, GLP 抑 glucagon

② 肠: 抑消化

③ 脑: 抑进食

④ 其他: GLP 提高心输出, GIP 促脂肪积累

Diabetes

症状: { ^{多尿} glycosuria: ^{渴 (多饮)} polyuria, ^{由 cell shrinking 引发} polydipsia, brain damage
intracellular glucose deficiency: polyphagia
Blood FAT/ketosis/metabolic acidosis
Blood aa T/muscle protein ↓
prolonged hyperglycemia ⇒ chronic skin infection ...

治疗: 口服降糖药 oral hypoglycaemic drug

- ① sulphonylurea: 磺脲. 抑制ATP-K⁺通道来促 insulin
- ② Glitazone (TZD): increase Glc uptake in adipose.
- ③ Metformin: 抑肝排 Glc, 促肌用 Glc.
- ④ incretin

Exercise: 提高 GLUT4 表达?

5. 其余激素

catecholamine 儿茶酚胺: 促糖异生, 糖原降解, lipolysis
压力下提高反应力

GH: 抑肌肉用糖, 促肝糖原分解, lipolysis, 促蛋白合成

GC: 升糖, lipolysis, 促肌 Protein 分解